

significant drawback of our educational system. According to Kapuria, project-driven learning, supported by a steady exposure to seminars on contemporary topics and interaction with the industry people, gives aspirants an edge. "A robotics-based project in the areas of different assistive systems may be a turnkey to enter a good organisation. Any parallel course on robotics may also be helpful. But nearly all our institutes, barring a handful, are woefully inept in providing students with such opportunities," he adds.

Major recruiters, however, often complain that they spend their quality time and resources in preparing freshers for specific job needs. This 'gap' between industry and institutes can be filled by taking up industry-oriented projects and strategically chosen courses run by various educational/technical centres.

Such training can go a long way in providing an edge over your competitors. So, concentrate on your project work and try to identify a course. These steps can increase your chances to be picked up. The recruiters need something extra, and the hands-on experience is indispensable.

You can get the overview of 'chip to ship' of robotics only after completing a project. Instead of depending only on your institutional projects, try for an industrial project or internship. Various private and public sector organisations accommodate final year and pre-final year students in their live projects.

We have already discussed the

Start-up speak

Job at first and then entrepreneurship, for a simple reason that hands-on experience is very important in this field. If a software engineer makes a mistake it could result in an error pop-up but if an electronics engineer makes a mistake it could lead to losses in terms of money and even physical injuries.

Not only R&D experience is necessary but also an exposure to technical marketing and procurement process can be helpful if you plan to start your own company. Contacts and business relations made once always pay back. Procurement of electronic/mechanical components is a very tricky job. As for the same functionality there are numerous alternatives, different manufactures and quality grades, it requires subjective awareness and experience.

After 1.5 years of service as R&D engineer, I founded TECHNIDO (Technology Nest in Italian) to work in industrial/educational robotics and technical training domains that was my childhood dream.



Siddharth Dev, chief technical officer & founder, Technido

relevant primary to-do steps. Besides these, you may try to understand project requirements in a better way with respect to design and documentation, development of prototype modules, coding and unit testing.

So be a go-getter, dig all the possibilities to get a practical exposure. Industry is ready to help you out. "I have delivered technical talks and demonstrated legged and wheeled mobile robots in various technical festivals and institutions. These activities have provided a way to students/aspirants to look beyond their normal curriculum and improve their approach towards robotics. We have handpicked the employees through personal interactions over the last four to five years. I am in touch with a number of students/researchers through various social networking forums and email," informs Vadakkepatt.

At the same time, keep yourself updated about the latest technological trends. Newspapers, technical magazines, various technical seminars and last but not the least—trade fairs—may provide you with an opportunity to get an edge over your competitors.

Touch up jobs

You may find various training institutes mushrooming all around. But before you join any course, be sure to compare the course curriculum with the skill sets discussed so far. Besides, you can always rely on vocational

courses offered by the institutes of repute (see Institute Watch).

Till now, we had enough chin wagging with robotics. According to my perception, now you are ready to explore the field. So pull up your socks! Plan your schedule for the final year project in this field and find a suitable industry to pursue it.

I know, the next possible question is: 'How?' Search the website of the organisations, send requests for project work and put in honest efforts to grab every possible opportunity.

Dr Vadakkepatt has a tip for you. "Competitions have played an important role in pushing technologies, leading to a number of technology start-ups. The FIRA Robot World Cup and Congress Bangalore 2010 is expected to act as a catalyst in these directions where the advancements in robotics will be showcased. FIRA India aims to propagate and popularise robotics and international level robotic competitions across India," he says.

Taking part in such events (competitions, conferences) paves the way for improving various skills. For an example, taking up a joint/collaborative project can enhance your networking, communication and management skills. Perhaps, it will be your first interaction with robotics industry, but believe me it will be a fruitful one. ●

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Institute watch

Universities

- IIT, Kanpur
- University of Hyderabad
- Jadavpur University, Kolkata
- BITS, Pilani

Industrial Centres

- Kuka Training Centre, Pune
- HiTech Robotics Training Centre, Gurgaon

Training Centres

- EFY Tech Centre
- Entech
- Thinklabs